

VAHIDULLAH TAC

Schmidt Science Postdoctoral Fellow
Stanford University

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Research interests: My research develops physics-aware generative AI frameworks for materials discovery, with applications spanning from soft biological materials to foods and engineering materials.

EDUCATION

PhD **Purdue University**, West Lafayette, IN

2021 - 2024 *Mechanical Engineering*

Dissertation title: Machine Learning with Hard Constraints: Physics-Constrained Constitutive Models with Neural ODEs and Diffusion

MSc **Middle East Technical University**, Ankara, Turkey

2016 - 2019 *Aerospace Engineering*

Thesis title: Micromechanical modeling of carbon nanotube – polymer composites

BSc **Middle East Technical University**, Ankara, Turkey

2012 - 2016 *Aerospace Engineering*

RESEARCH

EXPERIENCE

Stanford University **Postdoctoral Scholar**

Mechanical Eng. *Advisor: Dr. Ellen Kuhl*

2024 - present

Initiated and led an independent research direction on generative AI for materials discovery, using food and sustainable protein design as a model system for broader questions in inverse design, scientific machine learning, and constrained optimization.

Conceived, designed, and executed an AI-for-burgers research program, including problem formulation, dataset construction, model development, design-generation pipeline, evaluation strategy, and manuscript leadership.

Developed preference-guided generative AI frameworks for high-dimensional material design spaces where physical properties, sustainability metrics, nutrition, and human liking jointly define performance.

Purdue University **Research Assistant**

Mechanical Eng. *Advisor: Dr. Adrian Buzanza Tepole*

2021 - 2024

Led theoretical and computational research in solid mechanics, scientific machine learning, and the intersection of the two. Introduced machine learning paradigms with *a priori* satisfaction of the relevant physics and implemented for modeling material behavior under hyperelasticity, viscoelasticity, and damage. Incorporated the data driven material models in commercial finite element solvers to enable rapid adoption in the industry. Developed uncertainty quantification methods for biological materials using generative AI, and data-driven methods for inverse characterization of heterogeneous hyperelastic materials.

Stanford University **Visiting PhD Student**

Mechanical Eng.

2024 - 2024 Conducted a 1.5-month grant-supported research visit in the group of Prof. Ellen Kuhl through Purdue's TRACER program. The visit supported collaborative research and scholarly exchange within Stanford's computational mechanics community.

Turkish Aerospace Industries (TAI) Structural Design Engineer

2018 - 2021 Designed and analyzed spacecraft structural components, including panels, inserts, and brackets, with attention to structural integrity, manufacturability, and mission requirements. Led the development of TAI's first structural panel with embedded active cooling, integrating mechanical design considerations with thermal management needs for aerospace applications.

TÜBİTAK Space Research Engineer

2017 - 2018 Designed and developed two engineering software tools supporting aircraft conceptual design and composite failure analysis. The work combined aerospace engineering principles with computational implementation, enabling more systematic design evaluation and structural assessment workflows.

PUBLICATIONS

Manuscripts Under Review

1. **V. Tac**, A. Koosis, and E. Kuhl, "Texture Independently Drives Liking in AI-Generated Alternative Protein Burgers," *Under review*, 2026.
2. **V. Tac** and E. Kuhl, "[Generative AI for material design: a mechanics perspective from burgers to matter](#)," *Under review*, 2026.
3. **V. Tac**, C. Gardner and E. Kuhl, "[Generative artificial intelligence creates delicious, sustainable, and nutritious burgers](#)," *Under review*, 2026.

Peer-Reviewed Journal Publications

1. **V. Tac**, A. Amiri-Hezaveh, M. K. Rausch, G. N. Bechtel, F. S. Costabal and A. B. Tepole, "[Fully data-driven inverse characterization of heterogeneous materials with hyper-network neural ODEs](#)," *npj Computational Materials*, 2026.
2. **V. Tac**, E. Kuhl, and A. B. Tepole, "[Data-driven continuum damage mechanics with built-in physics](#)," *Extreme Mechanics Letters*, 2024.
3. **V. Tac**, M. K. Rausch, I. Bilonis, F. S. Costabal and A. B. Tepole, "[Generative hyperelasticity with physics-informed probabilistic diffusion fields](#)," *Engineering with Computers*, 2024.
4. L. Nunez-Alvarez, J.K. Ledwon, S. Applebaum, B. Progri, T. Han, J. Laudo, **V. Tac**, A. K. Gosain and A. B. Tepole, "[Tissue expansion mitigates radiation-induced skin fibrosis in a porcine model](#)," *Acta Biomaterialia*, 2024.
5. **V. Tac**, K. Linka, F. S. Costabal, E. Kuhl and A. B. Tepole, "[Benchmarking physics-informed frameworks for data-driven hyperelasticity](#)," *Computational Mechanics*, 2024.
6. **V. Tac**, M. K. Rausch, F. S. Costabal and A. B. Tepole, "[Data-driven anisotropic finite viscoelasticity using neural ordinary differential equations](#)," *Computer Methods in Applied Mechanics and Engineering*, 2023.

7. **V. Tac**, F. S. Costabal, and A. B. Tepole, “[Data-driven tissue mechanics with polyconvex neural ordinary differential equations](#),” *Computer Methods in Applied Mechanics and Engineering*, 2022.
8. **V. Tac**, V. D. Sree, M. K. Rausch, and A. B. Tepole, “[Data-driven modeling of the mechanical behavior of anisotropic soft biological tissue](#),” *Engineering with Computers*, 2022.
9. Y. Leng, **V. Tac**, S. Calve, A. B. Tepole, “[Predicting the mechanical properties of biopolymer gels using neural networks trained on discrete fiber network data](#),” *Computer Methods in Applied Mechanics and Engineering*, 2021.
10. **V. Tac** and E. Gurses, “[Micromechanical modelling of carbon nanotube reinforced composite materials with a functionally graded interphase](#),” *Journal of Composite Materials*, 2019.

Book Chapters

1. **V. Tac** and A. B. Tepole, “[A modeler’s guide to soft tissue mechanics](#),” in *Comprehensive Mechanics of Materials*, V. Silberschmidt, Ed., Oxford: Elsevier, 2024.

Conference and Other Publications

1. **W. Taj** and D. Coker, “[Dynamic frictional sliding modes between two homogenous interfaces](#),” *IOP Conference Series: Materials Science and Engineering*, 2018
2. **W. Taj**, “Experimental determination of the stress intensity factor using photoelasticity,” *Matter*, 2015

AWARDS & HONORS

■ Schmidt Science Fellowship

Schmidt Sciences, 2024

Highly selective postdoctoral fellowship providing \$110,000/year in support. Awarded to 32 fellows selected from more than 400 nominees from leading institutions worldwide.

■ Travel for Collaborative Research Award

Purdue University, 2023

Competitive grant providing up to \$10,000 to support travel to a host institution for scientific collaboration. Awarded annually to approximately four Purdue graduate students; supported a 1.5-month research visit to Stanford University.

■ Future Faculty Symposium Travel Award

Society of Engineering Science, 2023

Travel award supporting participation in the Future Faculty Symposium at the Society of Engineering Science Annual Technical Meeting. Awarded to promising PhD students and postdoctoral scholars pursuing academic careers.

■ Robert J. Melosh Medal

Duke University, 2022

Awarded annually for the best student paper in computational solid mechanics; selected as the outstanding PhD student recipient.

■ Ward A. Lambert Teaching Fellowship

Purdue University, 2022

Teaching fellowship providing pedagogical training, mentorship, and the opportunity to serve as instructor of record for an undergraduate course. Enabled teaching *Basic Mechanics I: Statics* to 100+ undergraduate students.

- **Ben M. Hillberry Graduate Scholarship**

Purdue University, 2022

Annual scholarship recognizing one highly productive graduate student in mechanical engineering at Purdue University.

- **High Impact Publication Award**

TÜBİTAK, 2020

Award recognizing publications in high-impact journals; received for a publication in *Journal of Composite Materials* during master's studies.

- **Success Scholarship**

Türkiye Burslari, 2016

Awarded to outstanding international students in Turkish universities.

- **3rd Place, Individual Aircraft Design Competition**

American Institute of Aeronautics and Astronautics, 2016

- **1st Place, Engineering Day Poster Competition**

Middle East Technical University, 2016

- **Study Abroad Scholarship**

Ministry of Higher Education of Afghanistan, 2010

INVITED TALKS

- **Generative AI for discovery - assessing the taste and texture of AI-made burgers**

Machine Learning and AI in Constitutive Modeling of Complex Materials

17th World Congress on Computational Mechanics, 2026

- **Physics-constrained, data-driven modeling of elastic and inelastic material behavior**

Living Matter Lab

Stanford University, 2024

- **Generative hyperelasticity with physics-informed probabilistic diffusion fields**

Student Chapter Seminar Series

USACM (Online), 2024

- **Physics-constrained data-driven modeling of the mechanical behavior of soft materials**

Flex Lab Seminar Series

Purdue University, 2023

- **Automatically polyconvex strain energy functions with Neural ODEs**

Melosh Symposium

Duke University, 2022

TEACHING EXPERIENCE

Purdue University Lambert Teaching Fellow/Supervised Instructor

2023 *Basic Mechanics I: Statics*

Completed pedagogical training through the Ward A. Lambert Teaching Fellowship and taught selected course sessions under the mentorship of experienced faculty.

Spring 2023

Lecturer/Instructor of Record

Basic Mechanics I: Statics

Held full instructional responsibility for a course section of 100+ undergraduate students, including lectures, student support, assessment, and coordination of course activities.

Fall 2023

Teaching interests: Continuum mechanics, statics, scientific machine learning, AI-aided discovery, machine learning for engineers, constitutive materials modeling

MENTORSHIP EXPERIENCE

Stanford University First-Year Research Rotation Program, Spring 2025

2025 Mentee: Xinhao Quan, First-year Ph.D. student

First-Year Research Rotation Program, Fall 2025

Mentee: Ivana Chen, First-year Ph.D. student

Wu Tsai Performance Undergraduate Research Program, Fall 2025

Mentee: Alexandria Jones, Undergraduate student

OPEN-SOURCE CODE & DATA CONTRIBUTIONS

AI4Food Creator, Lead Developer and Contributor

2026 <https://github.com/LivingMatterLab/AI4Food>

Curated data, code, and examples demonstrating generative AI algorithms for food.

node_inv Creator & Lead Developer

2025 https://github.com/tajtac/node_inv

Software for inverse characterization of hyperelastic materials using machine learning.

1 fork

node_damage Creator & Lead Developer

2024 https://github.com/tajtac/node_damage

Software for training flexible machine-learning-based continuum damage models using neural ODEs.

3 stars

node_diffusion Creator & Lead Developer

2023

https://github.com/tajtac/node_diffusion

Software for uncertainty quantification in physics-constrained machine learning material models using diffusion, and generating physically plausible heterogeneous fields.

4 stars

nvisco Creator & Lead Developer

2023 <https://github.com/tajtac/nvisco>

Software for training and validating flexible machine-learning-based models for anisotropic finite viscoelastic material behavior.

2 forks

hypermodelcomp Creator & Lead Developer

2023 <https://github.com/tajtac/hypermodelcomp>

Software for training, validating, and benchmarking multiple machine-learning-based models for hyperelastic material behavior.

16 stars, 3 forks

Anisotropic_NN Creator & Lead Developer

_UMAT https://github.com/tajtac/Anisotropic_NN_UMAT

2023 Software for training and validating a deep-learning based model for hyperelastic material behavior. Accompanied by an implementation in Abaqus FEA using material subroutines.

23 stars, 7 forks

NODE Creator & Lead Developer

2022 https://github.com/tajtac/NODE_v2

<https://github.com/tajtac/NODE>

Software for training and validating a polyconvex hyperelastic material model based on neural ODEs. Accompanied by an implementation in Abaqus FEA using material subroutines.

11 stars, 4 forks

TECHNICAL SKILLS

Machine Learning Methods Deep learning, Generative modeling, Neural ODEs, Physics-informed and Physics-constrained machine learning

Machine Learning Tools Python, Jax, TensorFlow, PyTorch

Finite Element Modeling Abaqus, Ansys, in-house finite element packages, Jax-FEM

Languages English, Turkish, Persian, Turkmen, Uzbek, elementary Russian

SERVICE & OUTREACH

Peer Review Reviewer for *Computer Methods for Biomechanics and Biomedical Engineering*, *eBioMedicine*, *Engineering with Computers*, *Applied Mathematical Modelling*, and *Mechanics Research Communication*.

- Academic Service** Judge, *Summer Undergraduate Research Symposium*, Purdue University, 2023
Treasurer, *Afghan Student Association*, Purdue University, 2023-2024
- Outreach** Author, popular science articles for *InnovatED* magazine, *TUBITAK Bilim ve Teknik*, *The National Interest*.

REFERENCES

Ellen Kuhl, Ph.D.

Professor, Stanford University

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Relationship: Postdoctoral advisor

Adrian Buganza Tepole, Ph.D.

Associate Professor, Columbia University

ab6035@columbia.edu

Relationship: Ph.D. advisor

Francisco Sahli Costabal, Ph.D.

Assistant Professor, Pontifical Catholic University of Chile

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Relationship: Collaborator

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Professor, Purdue University

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Relationship: Collaborator